Project Title: Knight of the Site: Multi-Tier Phishing URL Detection System

1. **Problem Statement**
    a. Cloud-only phishing scanners add network latency and expose user browsing history, while static client-side models struggle to detect newly created zero-day phishing domains. This project, as required, builds a two-tier system that runs fast, privacy-preserving URL checks locally and offloads uncertain links to a cloud server for deep analysis and continual retraining.
2. **System Architecture**
    a. **Tier 1 (Mobile/Edge Client):** Extracts lightweight features and runs a quantized model for instant classification.
    b. **Tier 2 (Cloud Server):** A FastAPI and PostgreSQL backend ingests flagged URLs, extracts heavy network features, retrains a stronger classifier, and stores updated weights for deployment back to Tier 1.

This specific design minimizes data transfer costs, enables low latency, and supports continual learning.

3. **Tech Stack:** ONNX/TFLite, FastAPI, Scikit-Learn/PyTorch, PostgreSQL, Docker, Git
4. **Data:** Kaggle, PhishTank API
5. **Evaluation Metrics:** Precision, Recall, F1-Score, False Positive Rate, latency (ms), model size